Daily Implementation & Systems Architecture Report

A comprehensive audit of all engineering updates, frontend enhancements, AI micro-quiz infrastructure, GitHub session authentication sync, and responsive layout calibrations deployed today across the CareerPath AI platform.

21	50+	6,500+	100%
COMMITS DEPLOYED TODAY	FILES TOUCHED & REFINED	LINES ADDED / CALIBRATED	QUALITY GATE PASS (TESTS & GRAPH)

1 Executive Highlights of Today's Deployments

1. Multi-Model AI Skill Verification Micro-Quiz (Groq + Gemini + BYOK)

BRAND NEW FEATURE

- **Dynamic Question Generation:** Powered by **Groq Llama 3.3 70B** (200ms ultra-fast inference) and **Google Gemini 2.5 Flash** for pedagocial skill testing without credential inflation.
- **Zero-Friction Fallback:** 860+ lines of curated domain questions across Python, JS, React, Node, SQL, ML, and Cloud/DevOps when offline.
- **One-Time Verification Rule:** Persists `isQuizVerified: true` and `hasCompletedSkillVerification: true` on user account so students are never blocked or re-gated.

2. Auth-Based GitHub Repository Synchronization (Zero Username Prompt)

KEY USER REQUEST

- **Session-Driven Sync:** Replaced the manual username modal with `POST /api/auth/github/sync` using Bearer JWT. Resolves student GitHub identity directly from MongoDB session.
- **Automatic Skill Verification:** Inspects repo code languages and commits, granting `isCodeVerified: true` with glowing verified checkmark badges across Dashboard and Assessment.

3. 90% Viewport Container Layout & 2D Percentage Progress Gauge

UI REDESIGN

- **Clean 2D Progress Indicator:** Eliminated confusing 3D radial angle degrees (e.g. 90/180) in favor of high-contrast, accessible 0–100% SVG circular meters.
- **Space Utilization:** Re-engineered CSS layout to fill 90% container width with balanced 10% side margins, removing dead whitespace and fixing sticky navbar clearance.

2 Exact Places & Pages Modified Today

A. Assessment Page

`client/assessment.html & client/js/assessment.js`

- **Auto-Detect Skills via Auth:** Clicking "Auto-Detect Skills" directly triggers `window.GitHubAuth.syncRepos()` with zero modal popups.
- **Live Verified Badges:** Selected skills render real-time green checkmark tags `[✓ Code Verified]` and `[✓ Quiz Verified]`.
- **TypeError Hardening:** Resolved runtime error `Cannot read properties of undefined (reading 'proficiency')` when selecting custom skills.
- **Popup Suppression:** Removed annoying auto-nudge modal popup so students can seamlessly complete their profile evaluation.
- **Responsive Width:** Container expanded to 90% with 10% margins; form controls and typography scaled up for readability.

B. Skill Verification Micro-Quiz Page (NEW)

`client/quiz.html, client/js/quiz.js, client/css/quiz.css`

- **Complete UI Flow:** Interactive 5-question micro-assessment with animated stepper dots, dynamic question countdown, radio options, and instant pedagogical answer explanations.
- **BYOK & Provider Switcher:** Built-in modal allowing evaluators to test live Groq, Gemini, or fallback generation with their own API keys.
- **Guest Auth Bridge:** Unauthenticated users can start the quiz in one click via automatic demo auth session seeding.
- **One-Time Persistence:** User profile updates in memory and MongoDB without re-gating subsequent assessments.

C. Student Dashboard

`client/dashboard.html, client/js/dashboard.js, client/css/dashboard.css`

- **One-Click Auth Repo Sync:** "Sync Repos" button uses the active session token; updates GitHub stats with inline loading spinners.
- **2D Percentage Circular Gauge:** Replaced WebGL 3D radial gauge with lightweight, responsive SVG meter showing exact percentage progress.
- **Full-Width Bento Grid:** Calibrated dashboard cards to 90% viewport width, eliminating awkward gaps and increasing font sizes.
- **Verified Skills Display:** Repositories analyzed and skill competencies rendered with code-verified pill badges.

D. Career Recommendations & Dynamic Roadmap Pages

`client/recommendations.html & client/roadmap.html`

- **Skill Label Visibility Fix:** Restored text visibility and eliminated label clipping on recommendation skill cards.
- **Skill-Gap Matrix:** Clearly marks matched skills, weak skills, and missing skills with verified code badges.
- **Roadmap Task Stepper:** Expanded weekly roadmap checklist to 90% width; completed tasks reflect percentage progress instantly.

3 Backend Services, APIs & Data Models

1. Dynamic AI Quiz Engine (Groq + Gemini + Static Fallback)

`server/services/aiQuizGeneratorService.js & quizService.js`

- **High-Speed Groq Integration:** Leverages `llama-3.3-70b-versatile` for sub-second question generation with strict JSON schema validation.
- **Gemini Multi-Modal Fallback:** Connects to Google Gemini 2.5 Flash for high-context technical reasoning questions.
- **Domain Bank:** 867-line curated fallback repository covering core competencies when offline.

2. Auth-Based GitHub Synchronization Controller

`server/routes/authRoutes.js & server/controllers/authController.js`

- **Route:** `POST /api/auth/github/sync` (Protected via JWT Bearer Token).
- **Identity Resolution Hierarchy:**

```
1. req.user.githubProfile.username → 2. req.user.githubId → 3. req.user.email prefix
```

- **Skill Extraction:** Analyzes student repositories, extracts languages, and updates `user.skills` with `isCodeVerified: true` in MongoDB Atlas.

3. Server Process Hardening & Resilience

`server/server.js`

- **Process Safety Handlers:** Attached `process.on('unhandledRejection')` and `process.on('uncaughtException')` preventing process crashes on unhandled async events.
- **CORS & Security:** Retains full localhost access (ports 5000 & 5500) and Vercel/Render production domains.

4. MongoDB Schema Enhancements

`server/models/User.js`

- **hasCompletedSkillVerification:** Boolean flag enforcing one-time verification so students aren't re-gated.
- **githubRepos:** Structured collection storing analyzed repository names, stars, and primary languages.
- **quizScore & quizGaps:** Tracks historical quiz assessment results and identified skill gaps.

4 File-by-File Implementation Changelog

FILE PATH	SUBSYSTEM	TYPE	IMPLEMENTATION SUMMARY
`client/quiz.html`	Frontend UI	CREATED	Adaptive micro-quiz page, 5-question stepper, BYOK modal.
`client/js/quiz.js`	Client Logic	CREATED	Quiz controller, question timer, provider toggle, guest auth bridge.
`client/css/quiz.css`	Styling	CREATED	Card styles, animated steppers, option selection, result states.
`server/services/aiQuizGeneratorService.js`	AI Service	CREATED	Multi-model AI generation engine (Groq Llama + Gemini Flash).
`server/controllers/quizController.js`	Backend API	CREATED	Quiz start, answer scoring, status queries, and skill verification.
`server/data/quizQuestions.js`	Data Engine	CREATED	860+ lines curated fallback questions across 8 core domains.
`server/routes/authRoutes.js`	Backend Auth	UPDATED	Protected route `POST /api/auth/github/sync`.
`server/controllers/authController.js`	Backend Auth	UPDATED	Auth-based GitHub repo fetching, language extraction, and verification.
`client/js/github-auth.js`	Client Auth	UPDATED	Added `syncRepos()`; routes `connectGitHubAccount` via token.
`client/js/assessment.js`	Client Assessment	UPDATED	Fixed undefined proficiency bug; auto-detects skills; verified checkmarks.
`client/js/dashboard.js`	Client Dashboard	UPDATED	Integrated auth repo sync; 2D SVG percentage gauge; wide layout.
`client/css/tokens.css & base.css`	Design System	UPDATED	90% container width with 10% margins; scaled typography; fixed navbar.
`server/server.js`	Core Server	UPDATED	Process safety handlers for unhandled rejections and exceptions.
`server/models/User.js`	Database Schema	UPDATED	Added `hasCompletedSkillVerification` and `githubRepos`.

5 Quality Gate & Pre-Flight Verification

- **Automated Test Suite:** `npm test` (server check & seed validation) passed cleanly with exit code 0.
- **Codebase Knowledge Graph:** `graft build` compiled 357 nodes, 617 edges, and 84 cards across JavaScript, TypeScript, and TSX files.
- **Live Health Check:** `GET /api/health` returning `HTTP 200 OK` connected to MongoDB Atlas shard.
- **Local Dev Server Integrity:** Backend on `http://localhost:5000` and Frontend on `http://localhost:5500` fully active.